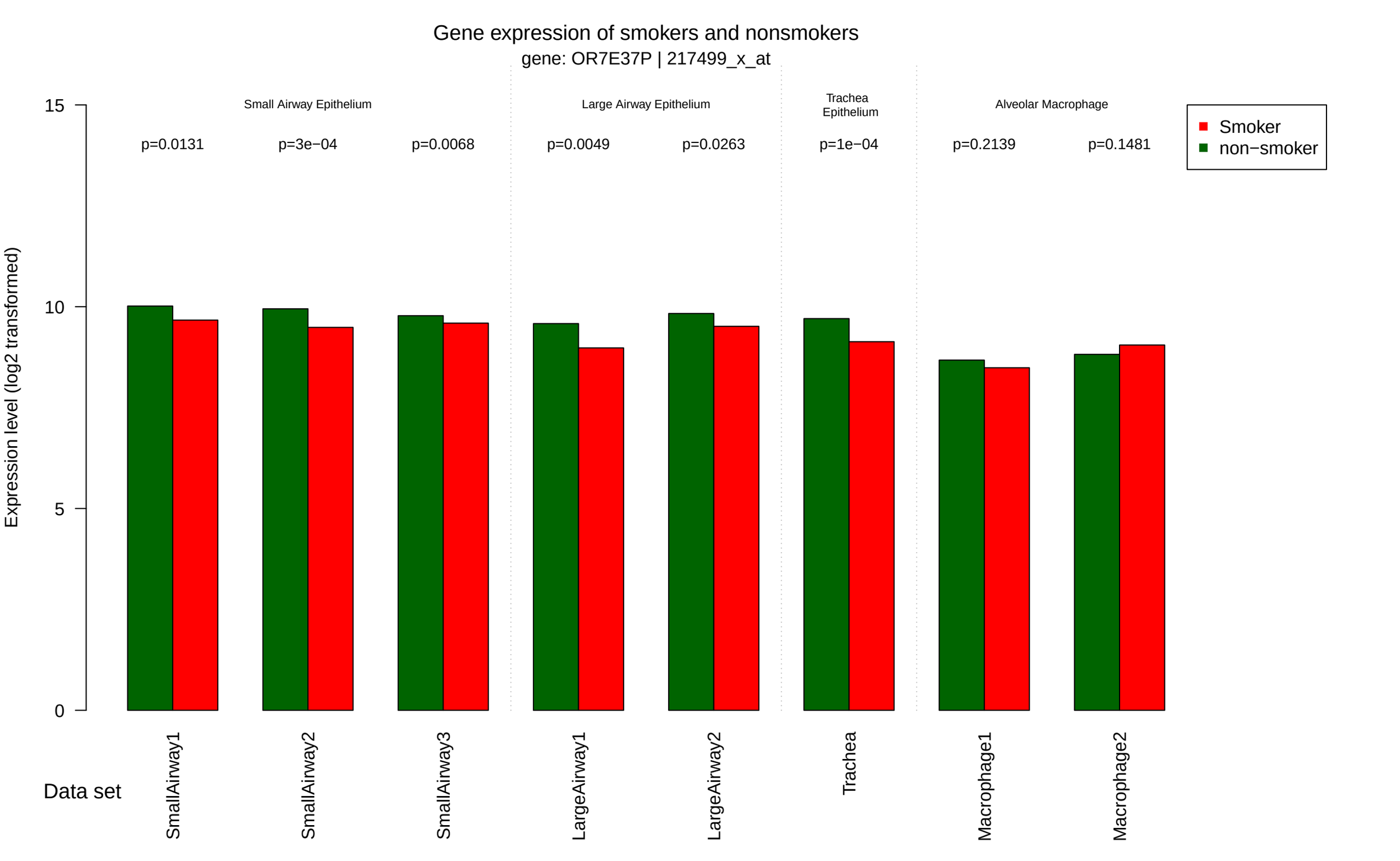
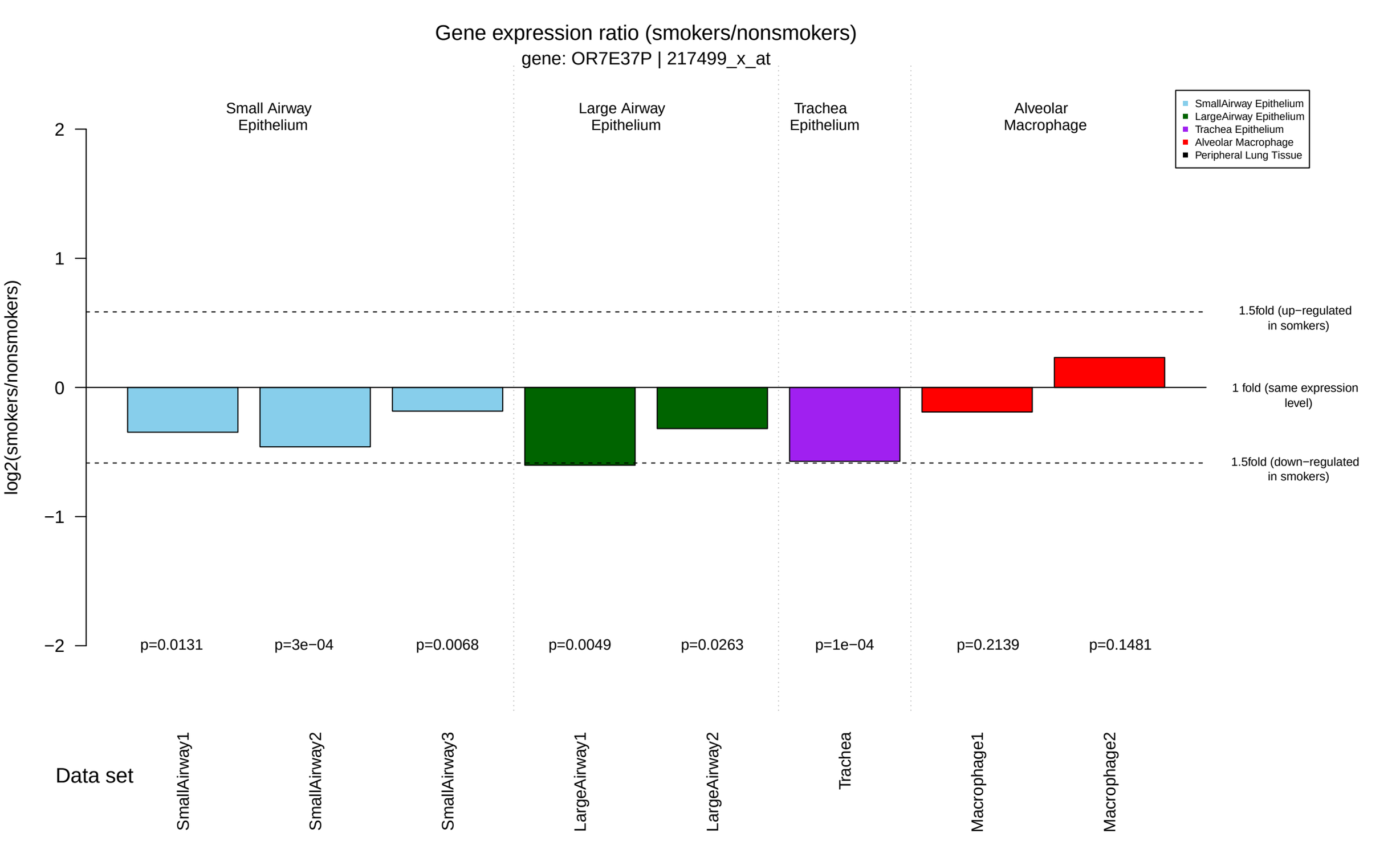
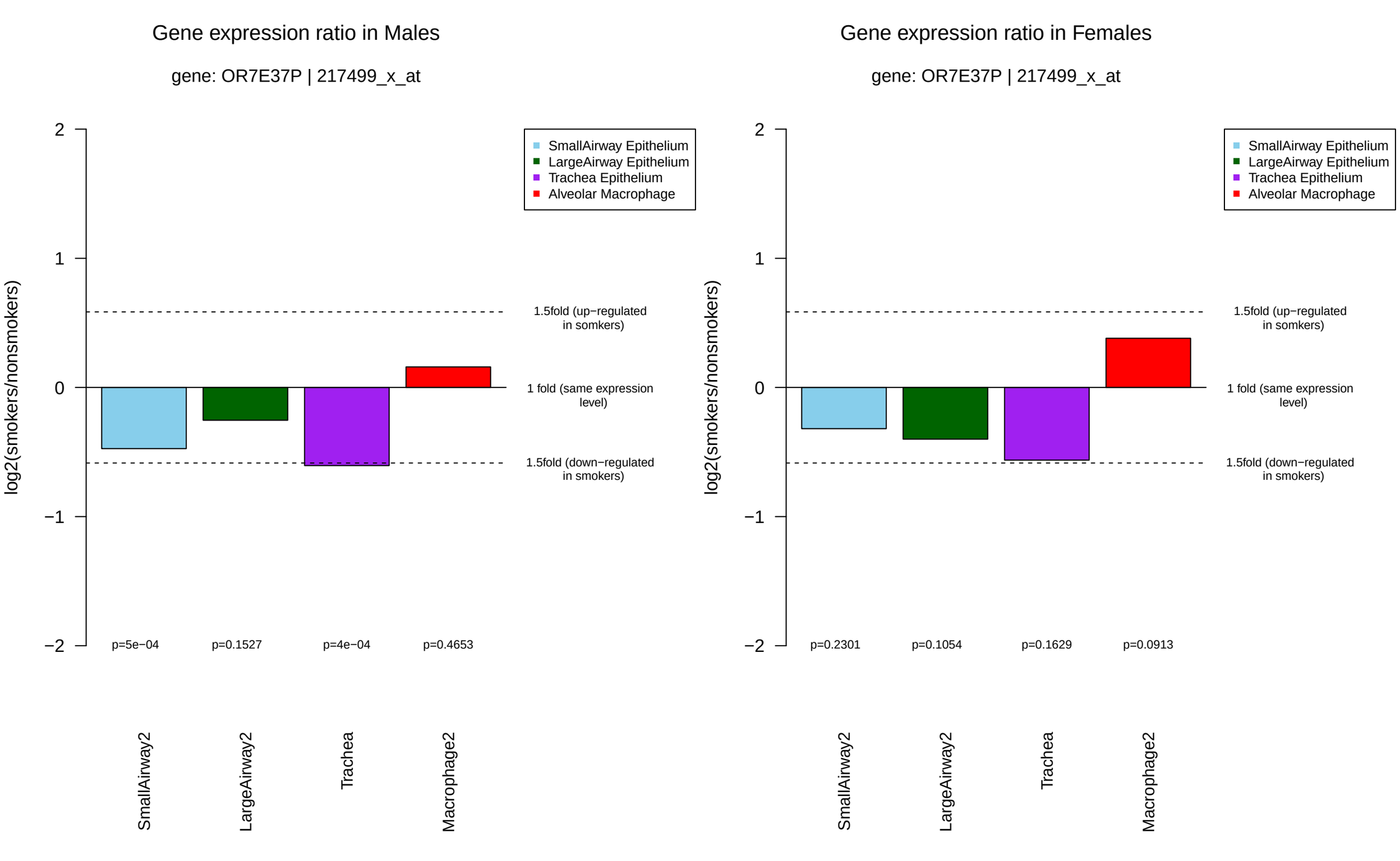
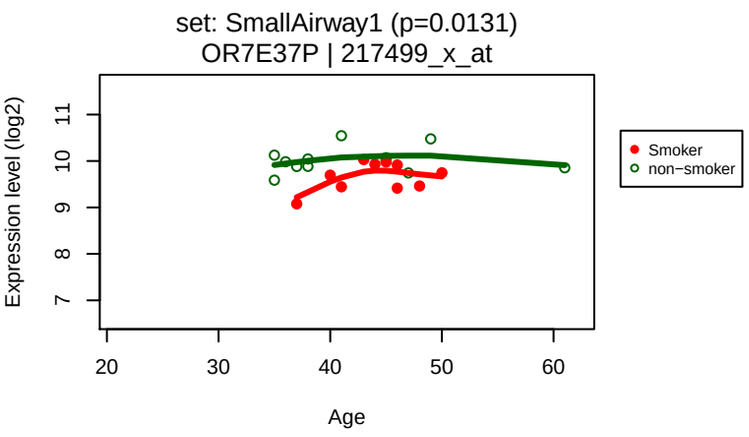
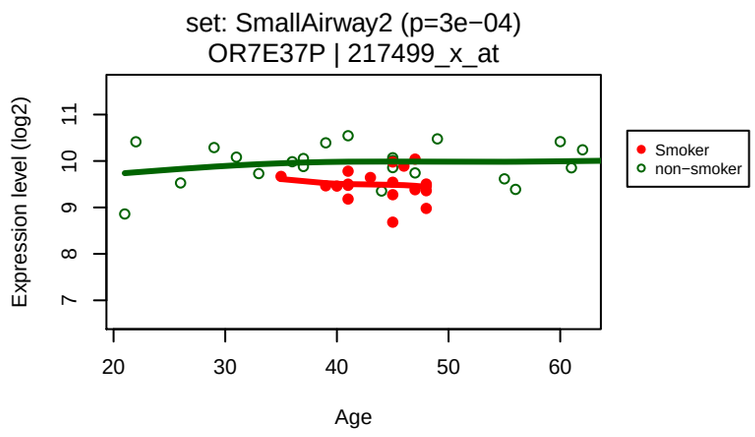
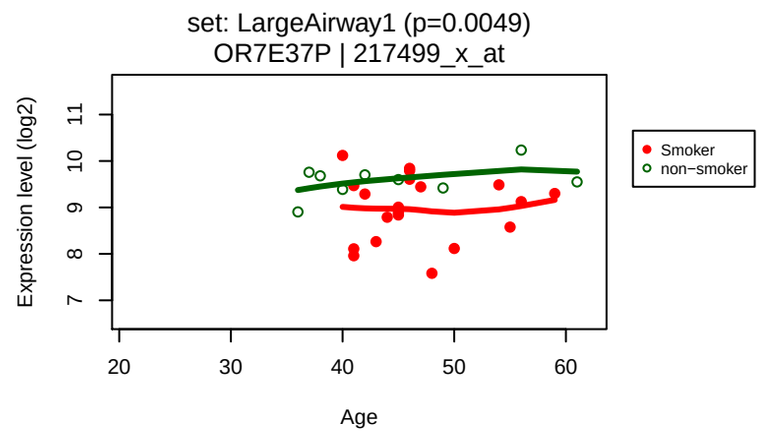

Gene expression of smokers and nonsmokers
gene: OR7E37P | 217499_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0131
p=3e−04
p=0.0068
p=0.0049
p=0.0263
p=1e−04
p=0.2139
p=0.1481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

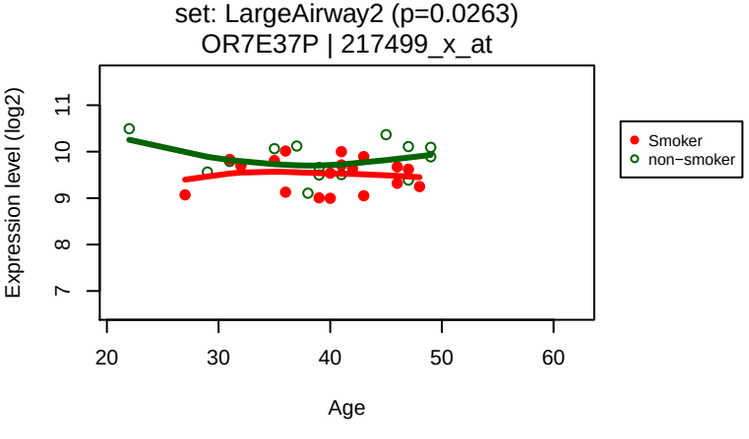

Gene expression ratio (smokers/nonsmokers)
gene: OR7E37P | 217499_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0131
p=3e−04
p=0.0068
p=0.0049
p=0.0263
p=1e−04
p=0.2139
p=0.1481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

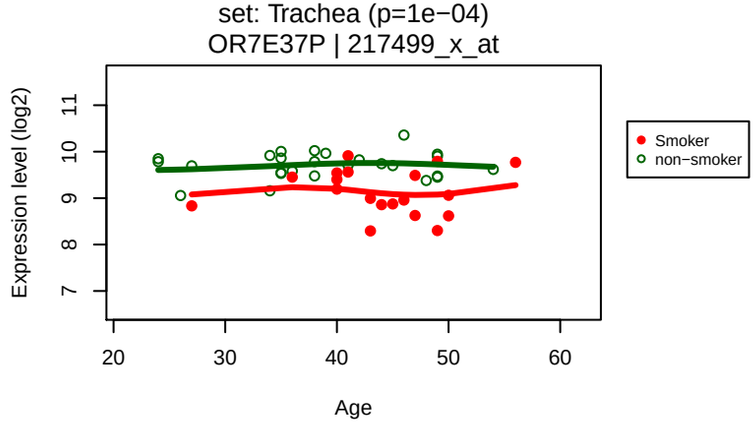

Gene expression ratio in Males
gene: OR7E37P | 217499_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.1527
p=4e−04
p=0.4653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR7E37P | 217499_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2301
p=0.1054
p=0.1629
p=0.0913
SmallAirway2
LargeAirway2
Trachea
Macrophage2

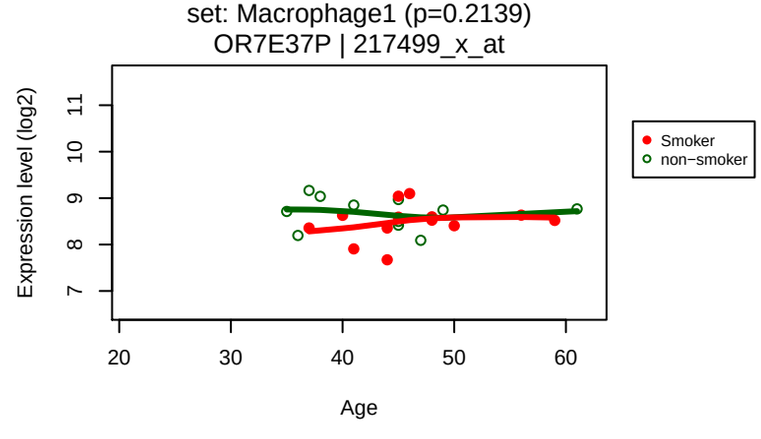

set: SmallAirway1 (p=0.0131)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker

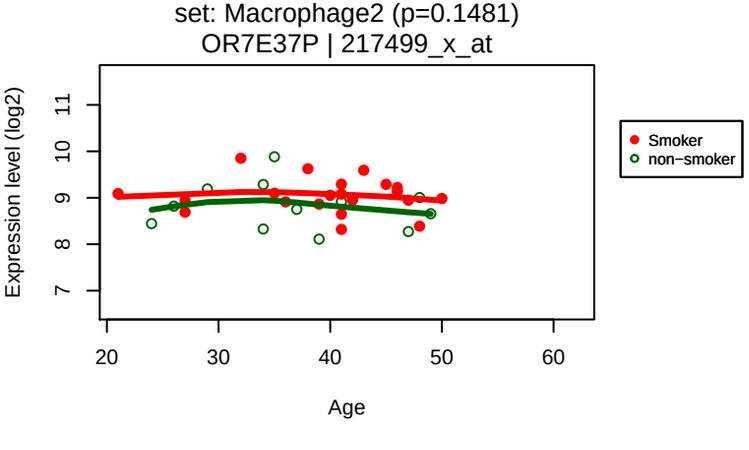

set: SmallAirway2 (p=3e−04)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker

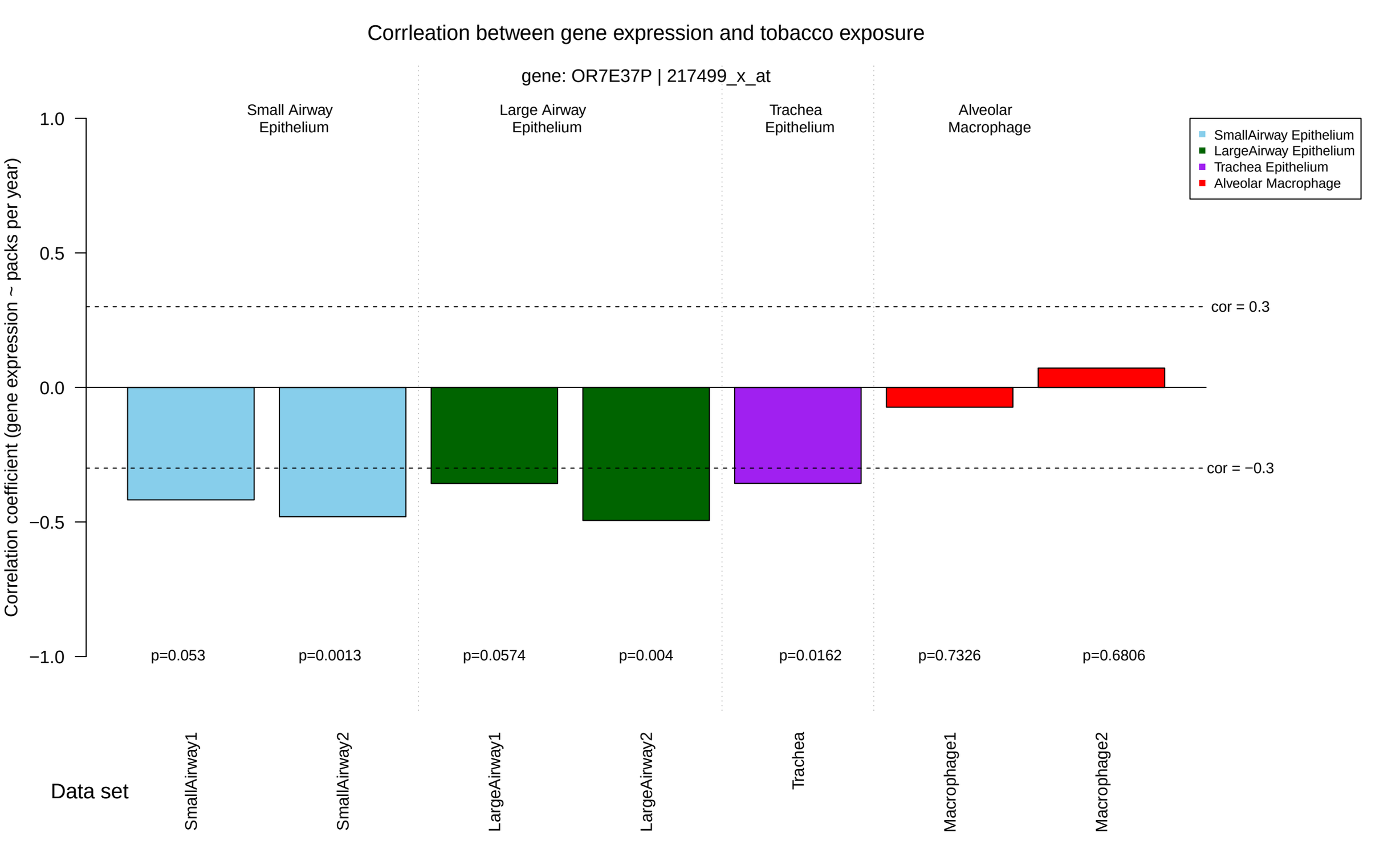

set: LargeAirway1 (p=0.0049)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0263)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2139)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1481)
OR7E37P | 217499_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR7E37P | 217499_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.053
p=0.0013
p=0.0574
p=0.004
p=0.0162
p=0.7326
p=0.6806
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2